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**Research Article**

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**Posted Date:** February 11th, 2026

**DOI:** https://doi.org/10.21203/rs.3.rs-7660029/v1

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**Additional Declarations:** No competing interests reported.

# ELEGANT: Combining Simultaneous Node and Edge Generation with Landmark Multi-Task Learning for Facial Action Unit Recognition

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**Abstract**

Facial Action Units (AUs) have recently been used in dementia detection, pain detection, talking head generation, and even facial reconstruction tasks. The success of each of these applications is at least partially due to the performance of the underlying AU recognition model. The commonly reported metric for comparing AU recognition is the average F1 score. Improving the average F1 score for AU recognition will directly improve the performance of each application that requires AU recognition. To improve the average F1 score, we propose simultaneously generating the nodes and edges for the graph neural network, strategically using landmark data, present additional AU recognition multi-task learning methods, and introduce ensemble learning to AU recognition. Although most current solutions for AU recognition generate the nodes and the edges separately, our proposed method demonstrates the improvement that comes from simultaneously generating the nodes and edges. In addition, our method proposes to use the available landmark data in a multi-task learning method. Our solution also applies ensemble learning to AU recognition. Through extensive experimentation, we demonstrate an improvement in the state-of-the-art average F1 score from 66.3 to 67.3 and from 66.9 to 67.8 on the BP4D and DISFA datasets, a considerable improvement in this field. These results underscore the substantial improvements our proposed method brings to the application of AU recognition.

**Keywords:** Ensemble learning, Facial AU recognition, Facial landmarks, Graph neural network, Multi-task learning, Transformer

## 1 Introduction

Facial action units (AUs) have been used recently in dementia detection [1], pilot mental workload assessment [2], talking head generation [3], artificial expression synthetization [4], and other tasks [5]. Each of these applications takes AU recognition as the input for its task. The performance of the underlying AU recognition model drives at least part of the success for each of these applications. Improving the average F1 score for an AU recognition model will also enhance the performance of each of these applications.

Current solutions for AU recognition commonly use Graph Neural Networks (GNNs) [6–12].

| | Previous Methods | ELEGANT |
|---|---|---|
| **Pre-training** | ✓ | ✓ |
| **GNN** | ✓ | ✓ |
| **Attention** | ✓ | ✓ |
| **Multi-Dimensional Edges** | ✓ | ✓ |
| **Simultaneous Node & Edge Creation (AGG Module)** | ✗ | ✓ |
| **Landmark prediction as multi task learning (COAL Module)** | ✗ | ✓ |
| **Ensemble Learning** | ✗ | ✓ |

**Table 1 Contributions:** We contribute simultaneous node & edge creation through our AGG module, using landmark predictions in multi-task learning through our COAL module, and ensemble learning. Together, these contributions form the ELEGANT solution.

Now that many use GNNs to improve AU recognition performance, the discussion has transitioned to focus on the method that generates the optimal nodes and edges. Current methods generate the nodes and edges separately [6–12] with some methods even generating the nodes and edges in separate training runs [11]. In addition, both the BP4D [13, 14] and DISFA [15, 16] datasets contain facial landmarks. Methods use these landmarks as an intermediate prediction to crop to certain areas of the face.

Despite the success achieved by generating nodes and edges separately, there remains potential for improvement in simultaneously creating both nodes and edges. A similar argument is found in related fields through model fusion or model consolidation. Model consolidation argues for better performance by merging multiple models into one model [17]. The reasoning behind model consolidation is that when a model learns multiple tasks, it can improve the overall performance on each task individually. The same reasoning can be applied to learning the nodes and edges simultaneously during training. In addition, many of the methods that do not crop to certain areas of the face throw away the landmark data. Ignoring the available data limits the overall performance of a system.

The current approach of generating nodes and edges separately is in contrast to the benefits found in model consolidation. In addition, the current use of landmark data limits which models can use this additional training data. We propose a solution that simultaneously learns the nodes and edges for the GNN and strategically uses the landmark data. We name our solution **E**nsemble **L**earning **E**ncoder **G**r**A**ph **N**eural Multi-**T**ask Learning **(ELEGANT)**. The simultaneous node and edge creation happens with our Attention Graph Generator (AGG) module. Later in the model's architecture, the landmark data is used to introduce landmark prediction as a Multi-Task Learning (MTL) method in our COntrastive Au-co-prediction Landmark (COAL) module. In addition, ELEGANT combines two base models in an ensemble learning approach, which increases performance even further. Notably, ELEGANT outperformed all previous AU recognition solutions on the average F1 metric even without ensemble learning. ELEGANT's contributions are compared with previous methods in Table 1.

In summary, the contributions of this work include:

1. The proposal of a neural network architecture, ELEGANT, which simultaneously learns the nodes and edges within the AGG module, allowing for interaction before being passed into the GNN.

2. We introduce using landmark predictions as an MTL approach for AU recognition within our COAL learning module. This additional training data provides a more complete understanding of the face in general. Previous solutions could only use the additional landmark data when their model architecture directly used it.

3. We also introduce additional MTL learning methods in the COAL learning module and ensemble learning for AU recognition. Our solution uses established deep learning methodologies specifically applied to AU recognition in areas that have not been extensively explored.

4. Extensive experimentation that provides evidence that the ELEGANT solution outperforms all other current state-of-the-art methods on the generally used AU recognition datasets: BP4D [13, 14] and DISFA [15, 16].

The subsequent sections of this paper are organized as follows. Section 2 provides an overview of the necessary background, followed by Section 3 with the proposed methodology. Then, Section 4 presents the associated results. Each element of our model is analyzed through the ablation study given in Section 5, and we provide a discussion of our findings in Section 6. Finally, we conclude in Section 7.

**Fig. 1 Example images:** These images are taken from the BP4D-Spontaneous and DISFA datasets. The left two are from the BP4D dataset, while the right two are from the DISFA dataset.

| AU # | FACS name | BP4D | DISFA |
|---|---|---|---|
| 1 | Inner brow raiser | ✓ | ✓ |
| 2 | Outer brow raiser | ✓ | ✓ |
| 4 | Brow lowerer | ✓ | ✓ |
| 6 | Cheek raiser | ✓ | ✓ |
| 7 | Lid tightener | ✓ | ✗ |
| 9 | Nose wrinkler | ✗ | ✓ |
| 10 | Upper lip raiser | ✓ | ✗ |
| 12 | Lip corner puller | ✓ | ✓ |
| 14 | Dimpler | ✓ | ✗ |
| 15 | Lip corner depressor | ✓ | ✗ |
| 17 | Chin raiser | ✓ | ✗ |
| 23 | Lip tightener | ✓ | ✗ |
| 24 | Lip pressor | ✓ | ✗ |
| 25 | Lips part | ✗ | ✓ |
| 26 | Jaw drop | ✗ | ✓ |

**Table 2 FACS Descriptions: F**acial **A**ction **C**oding **S**ystem (FACS) terminology corresponds to the AU numbers within the BP4D and DISFA datasets. A checkmark (✓) indicates inclusion within the dataset, while a cross (✗) denotes exclusion.

# 2 Background

AUs were originally developed as part of the Facial Action Coding System (FACS) to describe human facial motion using anatomical movements [18]. Real-world applications first began with depression analysis [19], micro-expression recognition [20], emotion recognition [21–23], and have now been applied in various fields [5]. The BP4D [13, 14] and DISFA [15, 16] datasets were released in 2013 to serve as benchmarks for AU recognition. Example images are shown in Figure 1 while Table 2 lists the AUs in these datasets. The average F1 score is generally recognized as the best metric to compare performance across different research in this field [9–12, 24–29].

## 2.1 Landmarks in AU Recognition

Both the BP4D [13, 14] and DISFA [15, 16] datasets provide landmark data for training purposes. Some methods use these landmarks for cropping to areas of the face [30], normalization [25], pre-processing techniques [24], cropping [8], or an anchor for their predictions [12]. Many of these methods have improved the average F1 score by using the provided landmark labels.

However, an increasing number of studies do not use landmark information [9–11, 26, 29, 31]. Some claim that using landmarks is restrictive [29]. Many others claim that their method performed better without using this additional information, but seem to suggest that having this additional data would improve results. However, these researchers have no way of including the landmark data in their solution's architecture and simply must set these data aside. Our solution proposes using these landmark data in our COAL learning module in a way that can be applied to all future research projects.

## 2.2 Graph Neural Network with AU Recognition

GNN based AU recognition approaches began in 2018 [6], and many have continued to use GNNs in their AU recognition architecture [7–12]. GNN based AU recognition architectures have focused improvements on providing probailistic solutions [7], using the edges as correlation between AUs [8], addressing class imbalance [9], and moving beyond maximum likelihood estimation [10]. Recent improvements have focused on making the edges in a GNN multidimensional [11] and combining region and global feature learning [12].

These methods generate the nodes and the edges in different locations within the network. Some methods, such as ME-GraphAU, generate the nodes and the edges even in separate training runs [11]. ELEGANT builds on previous advancements by simultaneously generating edges and nodes while uniquely incorporating established deep learning methodologies, offering a higher average F1 score in AU recognition.

## 2.3 Ensemble Learning

Ensemble learning, combines multiple models to improve performance. The first deep ensemble appeared in 2016, merging Bayesian and non-Bayesian neural networks [32]. Despite ensemble learning's long history, ELEGANT is the first AU recognition model to employ ensemble learning, leveraging its benefits to enhance AU recognition performance.

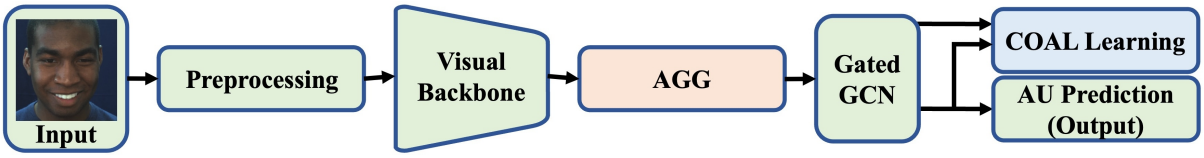


**Fig. 2 ELEGANT Base Learner Architecture:** The ELEGANT system presents unique advancements within the AGG (**A**ttention **G**raph **G**enerator) module and the COAL (**CO**ntrastive **A**U-Co-Prediction **L**andmark) learning module. We highlight the AGG module in red and the COAL module in blue. The portions of the architecture that are found in prior works are highlighted in green.

# 3 Methods

In this paper, we propose a new architecture, ELEGANT for AU recognition. In our approach, the image is passed through pre-processing, the visual backbone, and then our AGG module generates the nodes and edges simultaneously. Next, the edges and nodes are passed to the GCN, which is then passed to both our COAL learning module and the AU final prediction head. The base learner of our ELEGANT solution is given in Figure 2.

To maintain the baseline and accurately compare with previous studies, ELEGANT also uses MTCNN [33] to crop and align the face as pre-processing. The visual backbone is a ViT [34], specifically a Swin Transformer base model [35]. The graph neural network is a gated GCN [36]. As these three items are used in previous articles [11, 12], we do not give a further explanation in this article concerning these topics. However, we provide an in-depth discussion of our unique contributions to modifying and including the AGG, COAL learning, and Ensemble Learning modules for AU recognition.

## 3.1 AGG Module

While previous methods generate the nodes and edges independently [8–12], our method generates the nodes and edges simultaneously. This is accomplished by using attention as defined as:

$$Attention(Q, K, V) = softmax(\frac{QK^T}{\sqrt{d_k}})V \quad (1)$$

where Q is the query, K is the key, and V is the value as defined by Vaswani et al. [37]. Using the attention mechanism, the nodes and edges are generated simultaneously, allowing for each to learn from the other before any combination in the GNN.

Before the attention layers, we use a learned positional encoding that passes the input vector through a linear layer for each input, resulting in learning a unique representation for each positional location. Positional encodings are critical to attention heads. We have the total number of input and output nodes for the attention layers to be:

$$N = n \times n + n \quad (2)$$

where $N$ represents the number of inputs into the Encoder and $n$ represents the number of AUs being classified. This results in having enough outputs to then split the first $n$ outputs from the encoder to be the nodes and the remaining $n \times n$ elements to be the edges. We combine multiple encoder layers, using three for the BP4D dataset and five for the DISFA dataset. We provide a diagram to better visualize the AGG module in Figure 3. The output is then passed to the single-layer GNN.

## 3.2 COAL Learning

The **CO**ntrasitive **A**U-co-prediction **L**andmark (COAL) learning module is built to perform MTL and strategically use the landmark data included in the datasets. We include three additional tasks to the AU predictions to increase the overall performance: landmark prediction, contrastive embedding learning, and AU co-prediction. A schematic illustrating the COAL learning module is provided in Figure 4.

Including landmark predictions as an MTL method allows us to use the landmark data included in the datasets as additional training data. Using these data as training data allows us to combine the benefits of not being limited to

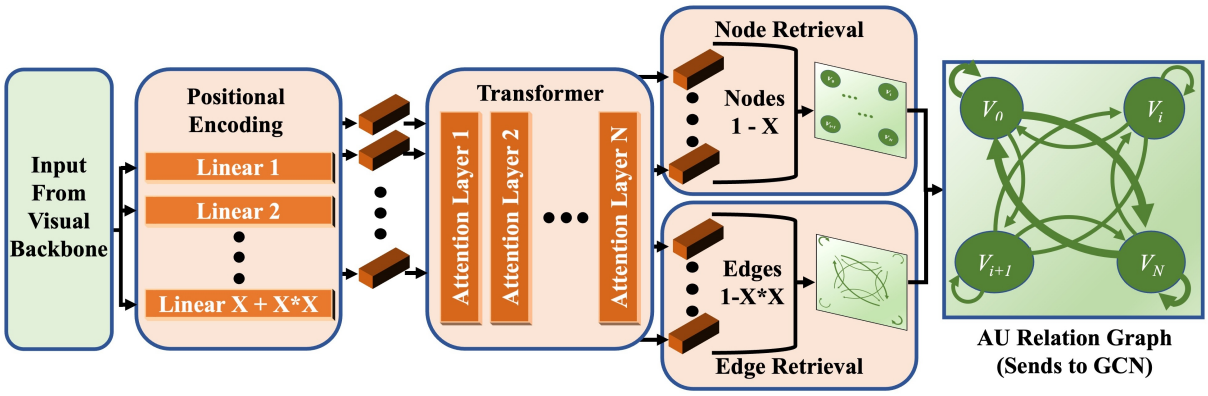


**Fig. 3 AGG Module:** While previous methods used linear or attention layers exclusively for edge generation [11], the AGG (Attention Graph Generator) module generates nodes and edges simultaneously before passing the data to the GNN. This ensures a more cohesive interaction between nodes and edges, not only in the GNN but also before it.

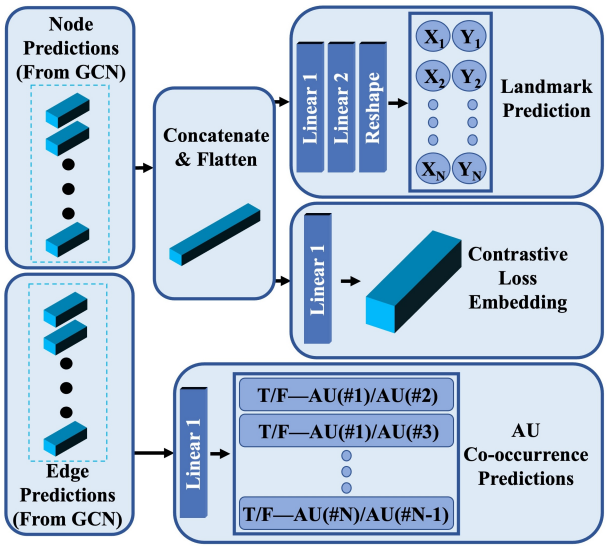


**Fig. 4 COAL Learning Module:** The COAL (**CO**ntrasitive **A**U-Co-Prediction **L**andmark) learning module enhances the average F1 score, as it engages the overall network in three supplementary tasks, which improves generalization. Specifically, the module employs landmark predictions, a contrastive loss embedding, and AU Co-occurrence predictions.

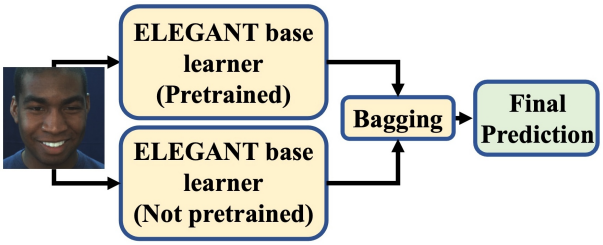


**Fig. 5 Ensemble Learning:** Ensemble learning is integrated into the ELEGANT approach through two networks. The initial network undergoes pre-training on the FEC dataset and subsequent fine-tuning for the AU recognition task, while the second network replaces the pre-training with random initialization. AU confidences between the two networks are combined in a bagging ensemble method to furnish a more accurate AU prediction.

cropping to landmark areas on the face as well as the benefit of additional training information.

## 3.3 Ensemble Learning

We note that using the FEC dataset for pre-training is beneficial for increasing the average F1 score for AU recognition. This is because pre-training distills specific information about facial structure into the network before the AU recognition training. However, the labels for the FEC dataset are simply between three images in which two facial expressions are more similar than the other facial expression. In this related task, the network can learn specific areas of the face with greater understanding. We find that it also hinders the network's ability to classify certain AUs due to the underlying bias in the FEC dataset.

We include the use or not use of the FEC dataset as the difference in the training data in our bagging ensemble approach, as seen in Figure 5. While researchers have previously pre-trained networks on the FEC dataset, we train two networks: one with the FEC pre-training weights and

one without. Then, we combine the two networks' final predictions in a bagging method.

### 3.4 Training Details

We first pretrain a base learner on the FEC dataset. We then train the AU recognition model with the COAL learning module, and then fine-tune the model directly on the AU prediction task without the COAL learning module.

For pretraining, we use the AdamW optimizer [38] with a learning rate weight decay of 5e-4. The pretraining is done on the FEC dataset task.

After pre-training, we train with the COAL learning module included. For our ensemble, we have one base model that uses the pretrained weights and another base model that uses randomly initialized weights. We use a learning rate of 1e-4 for both networks, regardless of their initialization method, across both datasets. We weight each loss from the COAL learning module by weights that were found by inspection, although we maintained the weights across folds to prevent overfitting. The initial training was conducted for 5 epochs, and the best-performing epoch was selected while ensuring that each fold used the same epoch to again prevent overfitting.

Upon completing the first training stage, we removed the COAL learning module, decreased the learning rate, and fine-tuned the model directly on the AU recognition task. We now train for 20 epochs, again selecting the same epoch across all folds to prevent overfitting. To maintain a valid comparison, we perform three-fold cross-validation using the same train and validation split as previous work [11, 12].

### 3.5 Data

**BP4D** [13] is a manually annotated dataset that includes facial videos in response to tasks that elicit different emotions. The dataset also contains ground truth 2D landmark points. The 41 participants pictured in the dataset ranged from 18 to 29 years old. Of the 41 participants, 23 were women and the remaining 18 were men. Six of the participants were Black, 11 were Asian, four were Hispanic or Latino, and 20 were White or Caucasian.

**DISFA** [15] is another manually annotated dataset that contains facial videos. The dataset also included 66 facial landmark points for each image. The dataset contains 27 participants aged 18 to 50 years old. 12 of the participants were women and 15 were men. One of the participants was Black, three were Asian, two were Hispanic or Latino, and 21 were White or Caucasian.

**FEC** [39] is not a AU recognition dataset. Rather, it is a dataset that contains triplets of images and includes a label for which image is the most different from the other two. The dataset is a large-scale facial expression dataset with 87,517 different images downloaded from Flickr. As this dataset does not contain any AU labels, we use it in the pre-training step of the ELEGANT method.

## 4 Results

We find that across the literature, not one solution dominates performance over all the selected AUs in the BP4D and DISFA datasets. This is seen with the improvements from line to line not fully outperforming all AUs in the previous line in Table 3 and Table 4. Rather, the focus is on the average F1 score across all AUs. Across AU recognition literature the average F1 score is used as the main metric for comparison [9–12, 24–29].

In Table 3, we compare against other AU recognition methods on the BP4D dataset. We find that the ELEGANT method obtains the best average F1 score on the BP4D dataset. Similarly, we compare the ELEGANT method against the same top-performing methods on the DISFA dataset in Table 4. The ELGEGANT method again outperforms all other methods that benchmark results with the average F1 score. Our ELEGANT solution achieves the highest average F1 score on both the BP4D and DISFA datasets.

## 5 Ablation Study

We provide an ablation study on the main components of our contributions. This is done to analyze how each contribution affects the overall performance individually and collectively.

### 5.1 AGG Module Ablation

The AGG module is central to the overall creation of the graph for the nodes and edges. As such, we cannot make comparison without it, as we will have no way to generate a graph. We use the prior work of ME-GraphAU to serve as a baseline. The

| Method | Individual AU F1 Scores | | | | | | | | | | | | Avg. F1 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | **1** | **2** | **4** | **6** | **7** | **10** | **12** | **14** | **15** | **17** | **23** | **24** | **Score (↑)** |
| DRML [40] | 36.4 | 41.8 | 43.0 | 55.0 | 67.0 | 66.3 | 65.8 | 54.1 | 33.2 | 48.0 | 31.7 | 30.0 | 48.3 |
| EAC-Net [30] | 39.1 | 35.2 | 48.6 | 76.1 | 72.9 | 81.9 | 86.2 | 58.8 | 7.5 | 59.1 | 35.9 | 35.8 | 55.9 |
| JAA-Net [24] | 47.2 | 44.0 | 54.9 | 77.5 | 74.6 | 84.0 | 86.9 | 61.9 | 43.6 | 60.3 | 42.7 | 41.9 | 60.0 |
| LP-Net [25] | 43.4 | 38.0 | 54.2 | 77.1 | 76.7 | 83.8 | 87.2 | 63.3 | 45.3 | 60.5 | 48.1 | 54.2 | 61.0 |
| ARL [31] | 45.8 | 39.8 | 55.1 | 75.7 | 77.2 | 82.3 | 86.6 | 58.8 | 47.6 | 62.1 | 47.4 | 55.4 | 61.1 |
| AU-GCN [8] | 46.8 | 38.5 | 60.1 | [80.1] | 79.5 | 84.8 | 88.0 | 67.3 | 52.0 | 63.2 | 40.9 | 52.8 | 62.8 |
| SRERL [7] | 46.9 | 45.3 | 55.6 | 77.1 | 78.4 | 83.5 | 87.6 | 63.9 | 52.2 | 63.9 | 47.1 | 53.3 | 62.9 |
| UGN-B [9] | 54.2 | 46.4 | 56.8 | 76.2 | 76.7 | 82.4 | 86.1 | 64.7 | 51.2 | 63.1 | 48.5 | 53.6 | 63.3 |
| HMP-PS [10] | 53.1 | 46.1 | 56.0 | 76.5 | 76.9 | 82.1 | 86.4 | 64.8 | 51.5 | 63.0 | 49.9 | 54.5 | 63.4 |
| SEV-Net [41] | **[58.2]** | [50.4] | 58.3 | **[81.9]** | 73.9 | **[87.8]** | 87.5 | 61.6 | 52.6 | 62.2 | 44.6 | 47.6 | 63.9 |
| AUFM [26] | 57.4 | **[52.6]** | [64.6] | 79.3 | **[81.5]** | 82.7 | 85.6 | 67.9 | 47.3 | 58.0 | 47.0 | 44.9 | 64.1 |
| FAUDT [27] | 51.7 | 49.3 | 61.0 | 77.8 | 79.5 | 82.9 | 86.3 | 67.6 | 51.9 | 63.0 | 43.7 | [56.3] | 64.2 |
| KDSRL [28] | 53.3 | 47.4 | 56.2 | 79.4 | [80.7] | 85.1 | 89.0 | 67.4 | 55.9 | 61.9 | 48.5 | 49.0 | 64.5 |
| ME-GraphAU [11] | 52.7 | 44.3 | 60.9 | 79.9 | 80.1 | [85.3] | 89.2 | **[69.4]** | 55.4 | 64.4 | 49.8 | 55.1 | 65.5 |
| SACL [12] | 57.8 | 48.8 | 59.4 | 79.1 | 78.8 | 84.0 | 88.2 | 65.2 | [56.1] | 63.8 | [50.8] | 55.2 | 65.6 |
| GTLE-Net [29] | **[58.2]** | 48.7 | 61.5 | 78.7 | 79.2 | 84.2 | **[89.8]** | 66.3 | **[56.7]** | [64.8] | **[53.5]** | 53.6 | [66.3] |
| **ELEGANT (Ours)** | 57.4 | 50.1 | **[66.9]** | 79.2 | 80.4 | 84.9 | [89.5] | [68.9] | 55.2 | **[65.6]** | [50.8] | **[59.1]** | **[67.3]** |

**Table 3 F1 scores on the BP4D dataset:** The proposed ELEGANT method surpasses all other methods on the average F1 score. We focus on the average F1 score for comparison, as none of the previous methods achieves the highest across all individual AU F1 scores. For ease of comparison, the highest-performing value in each column is bolded and bracketed, and the second-highest-performing value is bracketed.

| Method | AU (F1 Score) | | | | | | | | Avg. F1 |
|---|---|---|---|---|---|---|---|---|---|
| | **1** | **2** | **4** | **6** | **9** | **12** | **25** | **26** | **Score (↑)** |
| DRML [40] | 17.3 | 17.7 | 37.4 | 29.0 | 10.7 | 37.7 | 38.5 | 20.1 | 26.1 |
| EAC-Net [30] | 41.5 | 26.4 | 66.4 | 50.7 | **[80.5]** | **[89.3]** | 88.9 | 15.6 | 48.5 |
| AU-GCN [8] | 32.3 | 19.5 | 55.7 | **[57.9]** | 61.4 | 62.7 | 90.9 | 60.0 | 55.0 |
| SRERL [7] | 45.7 | 47.8 | 59.6 | 47.1 | 45.6 | 76.5 | 84.3 | 43.6 | 55.9 |
| JAA-Net [24] | 43.7 | 46.2 | 56.0 | 41.4 | 44.7 | 69.6 | 88.3 | 58.4 | 56.0 |
| LP-Net [25] | 29.9 | 24.7 | 72.7 | 46.8 | 49.6 | 72.9 | 93.8 | 65.0 | 56.9 |
| AUFM [26] | 41.5 | 44.9 | 60.3 | 51.5 | 50.3 | 70.4 | 91.3 | 55.3 | 58.2 |
| ARL [31] | 43.9 | 42.1 | 63.6 | 41.8 | 40.0 | 76.2 | 95.2 | 66.8 | 58.7 |
| SEV-Net [41] | 55.3 | 53.1 | 61.5 | 53.6 | 38.2 | 71.6 | **[95.7]** | 41.5 | 58.8 |
| UGN-B [9] | 43.3 | 48.1 | 63.4 | 49.5 | 48.2 | 72.9 | 90.8 | 59.0 | 60.0 |
| HMP-PS [10] | 38.0 | 45.9 | 65.2 | 50.9 | 50.8 | 76.0 | 93.3 | [67.6] | 61.0 |
| FAUDT [27] | 46.1 | 48.6 | 72.8 | [56.7] | 50.0 | 72.1 | 90.8 | 55.4 | 61.5 |
| ME-GraphAU [11] | 54.6 | 47.1 | 72.9 | 54.0 | [55.7] | 76.7 | 91.1 | 53.0 | 63.1 |
| KDSRL [28] | 60.4 | 59.2 | 67.5 | 52.7 | 51.5 | 76.1 | 91.3 | 57.7 | 64.5 |
| SACL [12] | 62.0 | **[65.7]** | **[74.5]** | 53.2 | 43.1 | [76.9] | [95.6] | 53.1 | 65.5 |
| GTLE-Net [29] | [64.5] | 63.2 | 70.1 | 47.7 | 53.6 | 76.2 | 94.8 | 65.1 | [66.9] |
| **ELEGANT (Ours)** | **[67.2]** | [64.1] | [74.2] | 52.2 | 47.0 | 73.1 | 95.1 | **[69.8]** | **[67.8]** |

**Table 4 F1 scores on the DISFA dataset:** The proposed ELEGANT method surpasses all other methods on the average F1 score. We focus on the average F1 score for comparison, as none of the previous methods achieves the highest across all individual AU F1 scores. The top-performing method in each column is bolded and bracketed, while the second top-performing method is bracketed.

most accurate comparison between ME-GraphAU and our method is seen in Models B, C, and D in Table 5 as ME-GraphAU has one MTL approach. The results on Models B, C, and D obtained average F1 scores of 65.4, 65.6, and 65.5, while the ME-Graph AU method achieved 63.1 on the DISFA dataset. The improvement of 2.3, 2.5, and 2.4 average F1 percentage points demonstrates the contributions of the AGG module.

## 5.2 COAL Learning Module Ablation

In the ELEGANT method, we include the landmark data in the final stage of the architecture, the COAL learning module. The baseline difference between including the COAL module increases the average F1 score by 2.3 average F1 score points as seen in the difference between Model A and Model E in Table 5. This highlights the contribution of the COAL learning module. In addition, Models B, C, and D in Table 5 demonstrate the individual improvement of each element of the COAL learning module. COAL 1 represents the AU co-occurrence predictions, COAL 2 represents the contrastive learning, and COAL 3 represents the landmark predictions.

| Method | AGG | COAL 1 | COAL 2 | COAL 3 | Pre-train | Ensemble | Average F1 Score (↑) |
|---|---|---|---|---|---|---|---|
| Model A | ✓ | | | | | | 64.7 |
| Model B | ✓ | ✓ | | | | | 65.4 |
| Model C | ✓ | | ✓ | | | | 65.6 |
| Model D | ✓ | | | ✓ | | | 65.5 |
| Model E | ✓ | ✓ | ✓ | ✓ | | | 67.0 |
| Model F | ✓ | ✓ | ✓ | ✓ | ✓ | | 67.7 |
| ELEGANT | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | 67.8 |

**Table 5 Overall Ablation Study:** Each model (A-F) has each of the contributions of this paper applied one at a time to demonstrate their individual contributions. The final line (ELEGANT) demonstrates the overall improvement with all the contributions of this paper included. COAL 1 corresponds to AU co-occurrence predictions, COAL 2 involves contrastive learning, and COAL 3 pertains to landmark predictions. All assessments were conducted using the DISFA dataset.

| Method | BP4D Avg. F1 Score (↑) | DISFA Avg. F1 Score (↑) |
|---|---|---|
| DRML [40] | 48.3 | 26.1 |
| EAC-Net [30] | 55.9 | 48.5 |
| JAA-Net [24] | 60.0 | 56.0 |
| LP-Net [25] | 61.0 | 56.9 |
| ARL [31] | 61.1 | 58.7 |
| AU-GCN [12] | 62.8 | 55.0 |
| SRERL [7] | 62.9 | 55.9 |
| UGN-B [9] | 63.3 | 60.0 |
| HMP-PS [10] | 63.4 | 61.0 |
| FAUDT [27] | 64.2 | 61.5 |
| ME-GraphAU [11] | 65.5 | 63.1 |
| SACL [12] | [65.6] | [65.5] |
| **ELEGANT with no pre-train, no Ensemble (Ours)** | **[65.9]** | **[67.0]** |
| **ELEGANT with pre-train and Ensemble (Ours)** | **[67.3]** | **[67.8]** |

**Table 6 No Additional Data Model Comparison:** Even without using additional pre-training data, the ELEGANT method outperforms all prior approaches on the average F1 score that do not use pre-training data.

| Method | BP4D Avg. F1 Score (↑) | DISFA Avg. F1 Score (↑) |
|---|---|---|
| SEV-Net [41] | 63.9 | 58.8 |
| AUFM [26] | 64.1 | 58.2 |
| KDSRL [28] | 64.5 | 64.5 |
| GTLE-Net [29] | [66.3] | [66.9] |
| **ELEGANT with no Ensemble (Ours)** | **[66.6]** | **[67.7]** |
| **ELEGANT with Ensemble (Ours)** | **[67.3]** | **[67.8]** |

**Table 7 With Additional Data Model Comparison:** Even without using the ensemble learning technique, the ELEGANT method outperforms all prior approaches on the average F1 score that use additional data during training.

### 5.3 Ensemble Learning Ablation

We demonstrate that even without the ensemble learning, the ELEGANT method still improves over the previous methods on the average F1 score. When we use no additional data and no ensemble learning, the ELEGANT method outperforms previous methods, as seen in Table 6. Many researchers have included additional data in training, such as pretraining on the FEC dataset. We demonstrate in Table 7 that the ELEGANT method outperforms all these methods when pretrained on the FEC dataset and without ensemble learning.

## 6 Discussion

Improving the average F1 score for AU recognition can enhance the performance of certain downstream applications. We find that simultaneously generating the nodes and edges while using landmark data as additional training data within our COAL learning module improves the average F1 score. Additionally, we find that while pretraining on the FEC dataset increases the average F1 score, it hinders performance on certain AUs.

Many previous methods generate nodes and edges in separate locations in the network, and even across different training runs. We find that using our AGG module to generate the nodes and edges simultaneously in the same location within the network architecture improves the average F1 score. We hypothesize that this is due to the nodes and edges attending to each other before being passed into the GNN. Future researchers can apply this finding by exchanging their graph generation method with positional encodings and transformer layers, as we described in our AGG module.

Before this article, in AU recognition, landmark data was used only in intermediate steps, such as cropping to certain locations. We find that using landmark predictions as an additional MTL method in the COAL learning module also improves the average F1 score. Future AU recognition models can apply this finding by adding

an MTL head for landmark prediction, which will make the network learn a more comprehensive facial representation.

We discovered that while pretraining on the FEC dataset improved the average F1 score, it actually decreased the F1 score for certain AUs. AUs such as inner brow raiser, brow lowerer, chin raiser, and lip tightener decreased in performance when the model was pretrained on the FEC dataset. We hypothesize that this is due to the fact that these AUs being smaller facial expressions. The FEC dataset labels which of the three images contains the most different facial expressions. As such, these smaller facial motions were primarily overlooked as the labelers were instructed to focus on the larger differences in facial expressions. As the model is pretrained on this dataset, it is taught to ignore these smaller facial motions, making it more difficult to relearn them during model training. Future researchers can apply this finding as we did by using a stacking ensemble or other methods that take into account the tradeoffs of using the FEC dataset for pretraining.

We acknowledge that our solution does not account for variation in lighting, nor extreme head rotation. In addition, as with many face recognition datasets [42], our solution is not tested in variation across race, gender, and age. These limitations are primarily due to the BP4D and the DISFA datasets being collected in a controlled environment. Future research can focus on developing evaluation sets that contain this variation or designing solutions to address these directly. As these limitations are outside the scope of this article, we maintain our position on the importance of our contributions to the field of AU recognition while simultaneously encouraging future research to address these limitations.

Future researchers can apply the findings of this article by using attention layers for generating the nodes and edges simultaneously, as we do in our AGG module. Researchers can also add landmark prediction as a way to incorporate additional data in an MTL learning approach, as we do in our COAL learning module. Researchers may also be interested in adding more MTL learning approaches, such as contrastive learning and AU co-occurrence predictions. Finally, researchers can be intentional with their use of pre-training data and acknowledge the tradeoffs that come from using it.

# 7 Conclusion

In conclusion, this study presents an AU recognition solution, ELEGANT, which incorporates a new AGG module, a COAL learning module, and ensemble learning. This solution introduces a unique approach that simultaneously develops the edges and nodes for the AU recognition task. It also provides three MTL methods that contribute to learning the AU recognition task more broadly: AU co-occurrence prediction, contrastive learning, and facial landmark prediction. Finally, this article brings the benefits of ensemble learning to the AU recognition field. The experiments demonstrate the improvements from each of these contributions and ultimately achieve SOTA results on the BP4D and DISFA datasets by increasing the average F1 score by 1.0 and 0.9 percentage points, respectively. The code for this method is available at https://github.com/byu-rvl/ELEGANT.

# Declarations

**Acknowledgements:** This work is made possible by researchers at the University of Denver and Binghamton University who created and provided the DISFA and BP4D datasets.

**Author contributions:** Andrew Sumsion: Conceptualization, methodology, software, validation, formal analysis, investigation, data curation, manuscript writing, manuscript review and editing, and visualization. Dah-Jye Lee: Conceptualization, resources, manuscript review and editing, supervision, project administration, and funding acquisition. All authors reviewed and approved the final version of the manuscript.

**Competing interests:** The authors declare no competing interests

**Data availability:** A link to our code is provided within the manuscript.

**Funding:** The authors did not receive support from any organization for the submitted work.

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